Anastasia Loukeri
Robotics' Control - School of Informatics
Assignment 06.

Changes and Explanations for Assignment 06

Added Y-axis sliding capability: A standard URDF prismatic joint only allows translation along a single axis. To achieve 2D sliding along both the x and y axes, I added a new virtual/intermediate link (slider_x_link). The base link connects to this intermediate link via a prismatic joint moving on the x-axis (slider_x_joint), and the intermediate link connects to the main slider_link via a second prismatic joint moving on the y-axis (slider_y_joint).

Increased slider height: To raise the camera above the ground, I increased the z dimension of the slider_link's geometry box from 0.15 to 1.0. I also adjusted the z origin for the visual, collision, and inertial tags from 0.075 to 0.5 (which is exactly half of the new height) so the box rests correctly on the base.

Modified arm rotation and placement: The origin of the arm_joint was moved upwards to z="1.0" so it connects at the top of the newly expanded slider_link. To make the arm rotate downwards, I modified the upper and lower limits of the revolute joint. Since positive rotation around the -y axis points the arm "upwards", I set the limits to lower="$-pi/2$" and upper="0", allowing the arm to only swing downwards towards the base.

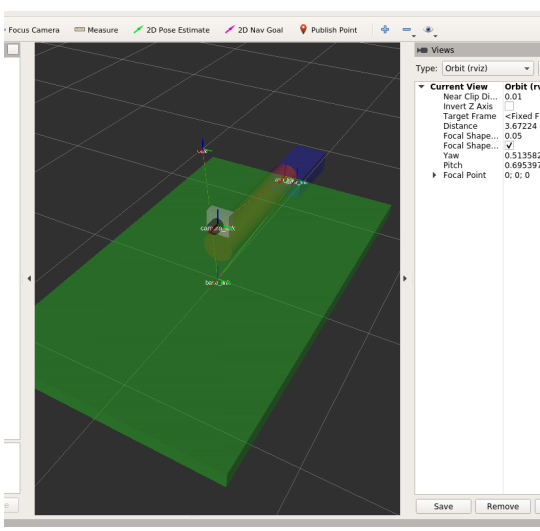

Figure 1: Initial Setup

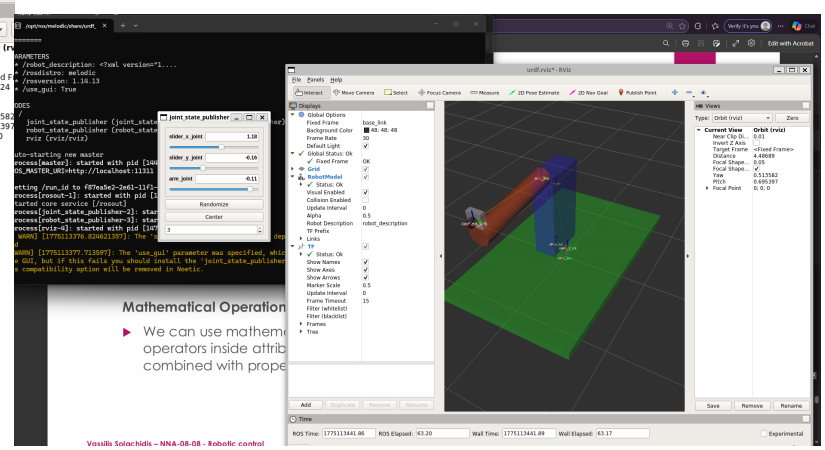

Figure 2: Modified Robot